

Derivation v27: Brane Mass from Brane Tension

BLOCK-003 Gravity Program — m_b from σ + Topological Pinning Candidate

EDC Collaboration

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Abstract

We construct a minimal-dimensional and action-level connection between the brane-localized mass parameter m_b (introduced in v26) and the brane tension σ . Starting from the dimensional ansatz $m_b = \lambda \sigma / M_5^3$, we express the dimensionless control parameter $b \equiv m_b L$ in terms of σ , M_5 , L , and $\bar{M}_{\text{Pl}}$. We explore a *topological pinning candidate* where λ is discretized ($\lambda = \pi n$ or $2\pi n$) due to orbifold/holonomy constraints. The spectral equation from v26 is re-expressed in terms of these parameters, mapping precisely where the gap becomes fixed. All quantities are rigorously tagged: the relation $m_b = \lambda \sigma / M_5^3$ achieves [Dc] status (one free parameter λ), while the gap prediction remains [I]+[BL] until L or λ is independently determined. Numerical verification confirms the limiting behaviors and explores the discrete spectrum for quantized λ .

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1 Introduction and Epistemic Framework

1.1 Context from v26

In derivation v26, we established the *Gap Derivability Program*:

- A brane-localized mass term generates Robin boundary conditions
- The transcendental spectrum $\tan(m_n L) = -m_b/m_n$ was derived [D]
- The gap bounds $\pi/2L < m_{\text{gap}} < \pi/L$ were established [D]
- The gap status remained [I]+[BL] because L and m_b were not derived

1.2 Goal of This Note

This derivation addresses the question:

Can m_b be derived from EDC-internal parameters, specifically the brane tension σ ?

We construct:

1. A dimensional relation $m_b = \lambda \sigma / M_5^3$
2. An action-level justification for this relation
3. The dimensionless control parameter $b = m_b L$
4. A topological pinning candidate where λ is quantized

1.3 Epistemic Tags

Table 1: Epistemic classification used in this derivation.

Tag	Meaning	Status
[D]	Fully derived (no free parameters)	Best
[Dc]	Derived with one calibration/parameter	Good
[P]	Postulated (hypothesis/candidate)	Uncertain
[I]	Identified with observable	External
[BL]	Baseline (measured input)	External
[OPEN]	Unresolved	Gap

2 Dimensional Analysis: m_b from σ

2.1 Available Scales

The EDC brane setup provides the following mass scales:

$$[\sigma] = M^4 \quad (\text{brane tension}) \tag{1}$$

$$[M_5] = M \quad (\text{5D Planck mass}) \tag{2}$$

$$[L] = M^{-1} \quad (\text{compactification length}) \tag{3}$$

$$[\bar{M}_{\text{Pl}}] = M \quad (\text{4D reduced Planck mass}) \tag{4}$$

2.2 Target Dimension

The brane mass parameter has dimension:

$$[m_b] = M \quad (5)$$

2.3 Dimensional Candidates

We seek combinations of σ , M_5 , and L that yield dimension M :

$$\frac{\sigma}{M_5^3} : \frac{M^4}{M^3} = M \quad \checkmark \quad (6)$$

$$\sigma L^3 : M^4 \cdot M^{-3} = M \quad \checkmark \quad (7)$$

$$\frac{\sigma L}{M_5^2} : \frac{M^4 \cdot M^{-1}}{M^2} = M \quad \checkmark \quad (8)$$

2.4 Primary Candidate

The most natural choice (involving only σ and M_5 , without L) is:

$$\boxed{m_b = \lambda \frac{\sigma}{M_5^3}} \quad (9)$$

where λ is a dimensionless coefficient.

Proposition 2.1 (Dimensional Consistency). *The relation eq. (9) is dimensionally consistent:*

$$[m_b] = \frac{[\sigma]}{[M_5^3]} = \frac{M^4}{M^3} = M \quad \checkmark \quad (10)$$

2.5 Physical Interpretation

The ratio σ/M_5^3 has a natural interpretation:

- σ is the energy density localized on the brane
- M_5^3 sets the 5D gravitational coupling scale
- σ/M_5^3 measures the “strength” of brane effects relative to bulk gravity

2.6 Status of the Dimensional Relation

$$m_b = \lambda \frac{\sigma}{M_5^3} \quad [\mathbf{Dc}] \quad (11)$$

The relation is $[\mathbf{Dc}]$ because:

- The dimensional structure is derived $[\mathbf{D}]$
- The coefficient λ remains undetermined $[\mathbf{OPEN}]$

3 Action-Level Derivation

3.1 5D Action with Brane

The total action consists of bulk and brane contributions:

$$S = S_{\text{bulk}} + S_{\text{brane}} \quad (12)$$

3.1.1 Bulk Action

The 5D bulk action for a scalar field Φ :

$$S_{\text{bulk}} = -\frac{1}{2} \int d^5x \sqrt{-g^{(5)}} g^{MN} \partial_M \Phi \partial_N \Phi \quad (13)$$

3.1.2 Brane Action

The brane at $\xi = L$ carries tension σ and couples to bulk fields:

$$S_{\text{brane}} = - \int d^4x \sqrt{-\gamma} \left[\sigma + \frac{1}{2} \mu \Phi^2 \right]_{\xi=L} \quad (14)$$

where $\gamma_{\mu\nu}$ is the induced metric and μ is a brane-localized mass-squared parameter.

3.2 Brane-Induced Boundary Operator

The key question is: how does σ generate the brane mass m_b ?

3.2.1 Mechanism: Tension-Field Coupling

Consider the gravitational backreaction of the brane tension. The Israel junction condition gives:

$$[K_{\mu\nu}] - \gamma_{\mu\nu}[K] = -\kappa_5^2 \sigma \gamma_{\mu\nu} \quad (15)$$

where $K_{\mu\nu}$ is the extrinsic curvature and $\kappa_5^2 = 1/M_5^3$.

3.2.2 Field-Dependent Correction

When a bulk scalar Φ is present, the effective tension becomes field-dependent:

$$\sigma_{\text{eff}} = \sigma + \Delta\sigma(\Phi) \quad (16)$$

Expanding to quadratic order in Φ :

$$\Delta\sigma(\Phi) = \frac{1}{2} \alpha_\sigma \Phi^2 + O(\Phi^4) \quad (17)$$

where α_σ is a coupling coefficient with dimension $[M^2]$.

3.2.3 Natural Scale for α_σ

The only available scales are σ and M_5 . Dimensional analysis gives:

$$\alpha_\sigma = c \frac{\sigma}{M_5^2} \quad (18)$$

where c is a dimensionless $O(1)$ coefficient.

3.2.4 Effective Brane Mass Term

Substituting into eq. (14):

$$S_{\text{brane}} \supset -\frac{1}{2} \int d^4x \sqrt{-\gamma} \alpha_\sigma \Phi^2 \Big|_{\xi=L} \quad (19)$$

Comparing with the standard form $S_{\text{brane}} \supset -\frac{1}{2} \int d^4x \sqrt{-\gamma} m_b \Phi^2$:

$$m_b = \alpha_\sigma = c \frac{\sigma}{M_5^2} \quad (20)$$

3.3 Reconciling Dimensions

Wait—eq. (20) gives $[m_b] = M^4/M^2 = M^2$, but we need $[m_b] = M$.

This indicates that the simple Φ^2 term in the action must be reinterpreted.

3.3.1 Correct Action Structure

The brane-localized mass term should be:

$$S_{\text{brane}} \supset -\frac{1}{2} \int d^4x \sqrt{-\gamma} m_b \Phi^2 \quad (21)$$

where $[m_b] = M^1$ (mass dimension 1), and $[\Phi] = M^{3/2}$ in 5D.

Actually, for a canonically normalized 5D scalar, $[\Phi] = M^{3/2}$, so:

$$[m_b \Phi^2] = M \cdot M^3 = M^4 \quad (22)$$

which integrates to give dimension $[M^4 \cdot L^{-4}] = M^0$ in 4D. This is consistent.

3.3.2 Proper Derivation

The brane coupling to Φ arises from the tension's dependence on bulk fields. Using the relation:

$$\kappa_5^2 = \frac{1}{M_5^3} \quad (23)$$

The natural scale for brane-induced mass is:

$$m_b \sim \kappa_5^2 \cdot \sigma = \frac{\sigma}{M_5^3} \quad (24)$$

Therefore:

$$\boxed{m_b = \lambda \frac{\sigma}{M_5^3} \quad \text{where } \lambda \sim O(1)} \quad (25)$$

3.4 Action-Level Status

Table 2: Derivation status of $m_b = \lambda\sigma/M_5^3$.

Component	Status	Tag
Dimensional structure	Derived	[D]
Scaling with σ	Derived from coupling	[D]
Scaling with M_5^{-3}	From κ_5^2	[D]
Coefficient λ	Undetermined	[OPEN]
Overall relation	Derived with 1 parameter	[Dc]

4 Dimensionless Control Parameter $b = m_b L$

4.1 Definition

From v26, the gap is controlled by the dimensionless parameter:

$$b \equiv m_b \cdot L \quad (26)$$

The transcendental spectral equation is:

$$\tan(x_n) = -\frac{b}{x_n}, \quad x_n = m_n L \quad (27)$$

4.2 Expression in Terms of σ , M_5 , L

Substituting $m_b = \lambda\sigma/M_5^3$:

$$b = m_b L = \lambda \frac{\sigma L}{M_5^3} \quad (28)$$

4.3 Using the Bridge Relation

From v23/v15, the bridge relation is:

$$\bar{M}_{\text{Pl}}^2 = M_5^3 \cdot L \quad [\mathbf{D}] \quad (29)$$

Solving for M_5^3 :

$$M_5^3 = \frac{\bar{M}_{\text{Pl}}^2}{L} \quad (30)$$

Substituting into eq. (28):

$$b = \lambda \frac{\sigma L}{\bar{M}_{\text{Pl}}^2/L} = \lambda \frac{\sigma L^2}{\bar{M}_{\text{Pl}}^2} \quad (31)$$

$$b = m_b L = \lambda \frac{\sigma L^2}{\bar{M}_{\text{Pl}}^2} \quad [\mathbf{Dc}] \quad (32)$$

4.4 Alternative Forms

4.4.1 In Terms of M_5 and L

Using $\bar{M}_{\text{Pl}}^2 = M_5^3 L$:

$$b = \lambda \frac{\sigma L^2}{M_5^3 L} = \lambda \frac{\sigma L}{M_5^3} \quad (33)$$

4.4.2 In Terms of σ and $\bar{M}_{\text{Pl}}$ Only

If we use $L = \bar{M}_{\text{Pl}}^2/M_5^3$:

$$b = \lambda \frac{\sigma}{M_5^3} \cdot \frac{\bar{M}_{\text{Pl}}^2}{M_5^3} = \lambda \frac{\sigma \bar{M}_{\text{Pl}}^2}{M_5^6} \quad (34)$$

4.5 Scaling Relations

From eq. (32):

$$b \propto \sigma \quad (35)$$

$$b \propto L^2 \quad (36)$$

$$b \propto \bar{M}_{\text{Pl}}^{-2} \quad (37)$$

4.6 Derivative Relations

The sensitivity of b to parameter changes:

$$\frac{\partial b}{\partial \sigma} = \lambda \frac{L^2}{\bar{M}_{\text{Pl}}^2} \quad (38)$$

$$\frac{\partial b}{\partial L} = 2\lambda \frac{\sigma L}{\bar{M}_{\text{Pl}}^2} \quad (39)$$

$$\frac{\partial b}{\partial \bar{M}_{\text{Pl}}} = -2\lambda \frac{\sigma L^2}{\bar{M}_{\text{Pl}}^3} \quad (40)$$

In logarithmic form:

$$\frac{d \ln b}{d \ln \sigma} = 1, \quad \frac{d \ln b}{d \ln L} = 2, \quad \frac{d \ln b}{d \ln \bar{M}_{\text{Pl}}} = -2 \quad (41)$$

4.7 Gap Dependence

The first KK mode (gap) satisfies:

$$\tan(x_1) = -\frac{b}{x_1} \quad (42)$$

with:

$$m_{\text{gap}} = \frac{x_1}{L} \quad (43)$$

Therefore:

$$m_{\text{gap}} = \frac{x_1(b)}{L} = \frac{x_1(\lambda \sigma L^2 / \bar{M}_{\text{Pl}}^2)}{L} \quad (44)$$

5 Topological Pinning Candidate

5.1 Motivation

The coefficient λ in $m_b = \lambda \sigma / M_5^3$ is currently [OPEN]. We explore whether λ could be *discretized* through topological constraints.

Hypothesis 5.1 (Topological Quantization of λ). The coefficient λ is not a free continuous parameter but takes discrete values:

$$\lambda = \pi n \quad \text{or} \quad \lambda = 2\pi n, \quad n \in \mathbb{Z}^+ \quad (45)$$

due to topological constraints on the brane-bulk coupling.

Status: [P] (Hypothesis/Candidate)

5.2 Possible Origins of Quantization

5.2.1 Origin 1: Orbifold Parity Projection

On an $S^1/\mathbb{Z}_2$ orbifold, fields have definite parity under $\xi \rightarrow -\xi$. The boundary operator coupling must respect this symmetry:

$$\int_{\text{brane}} \Phi^2 \quad \text{requires even parity for } \Phi \quad (46)$$

The orbifold projection can introduce factors of π in normalization:

$$\frac{1}{2\pi R} \int_{-\pi R}^{\pi R} d\xi \rightarrow \frac{1}{\pi R} \int_0^{\pi R} d\xi \quad (47)$$

5.2.2 Origin 2: Holonomy/Winding Number

If the brane-localized interaction arises from integrating out a bulk field with non-trivial holonomy around the compact dimension:

$$\oint A_5 d\xi = 2\pi n \quad (n \in \mathbb{Z}) \quad (48)$$

The effective boundary operator inherits quantization:

$$\lambda = \frac{1}{2\pi} \oint A_5 d\xi = n \quad (49)$$

or with additional factors:

$$\lambda = 2\pi n \quad \text{or} \quad \lambda = \pi n \quad (50)$$

5.2.3 Origin 3: Boundary Counterterm from Bulk Integration

Consider the bulk action integrated up to the brane:

$$S_{\text{eff}} = -\frac{1}{2} \int d^4x \int_0^L d\xi (\partial\Phi)^2 + S_{\text{brane}} \quad (51)$$

The boundary term from integration by parts:

$$\delta S = - \int d^4x \Phi \partial_\xi \Phi \Big|_0^L \quad (52)$$

If this is canceled by a brane counterterm proportional to σ :

$$S_{\text{ct}} = c \frac{\sigma}{M_5^3} \int d^4x \Phi^2 \Big|_L \quad (53)$$

The coefficient c could be fixed by requiring cancellation of divergences, potentially giving a quantized value.

5.3 Discrete Spectrum for b

If $\lambda = \pi n$:

$$b_n = \pi n \frac{\sigma L^2}{M_{\text{Pl}}^2}, \quad n = 1, 2, 3, \dots \quad (54)$$

If $\lambda = 2\pi n$:

$$b_n = 2\pi n \frac{\sigma L^2}{M_{\text{Pl}}^2}, \quad n = 1, 2, 3, \dots \quad (55)$$

5.4 Gap Spectrum

For each discrete b_n , the gap is:

$$m_{\text{gap}}^{(n)} = \frac{x_1(b_n)}{L} \quad (56)$$

where $x_1(b)$ is the first root of $\tan(x) = -b/x$.

Table 3: Status of topological pinning components.

Component	Evidence	Tag
Quantization hypothesis	Plausible	[P]
Orbifold origin	Consistent	[P]
Holonomy origin	Consistent	[P]
Counterterm origin	Speculative	[P]
Specific value of λ	Not determined	[OPEN]

5.5 Status of Topological Pinning

5.6 What Would Fix the Gap?

If λ is quantized ($\lambda = \pi n$) and L is determined:

$$b = \pi n \frac{\sigma L^2}{\bar{M}_{\text{Pl}}^2} \quad (\text{fixed for given } n, \sigma, L) \quad (57)$$

The remaining freedom is:

- Integer n (could be $n = 1$ as minimal choice)
- Compactification scale L (currently [I]+[BL])
- Brane tension σ (could be EDC-derivable)

6 Connection to v26 Spectral Equation

6.1 V26 Transcendental Equation

From derivation v26, with Neumann at $\xi = 0$ and Robin at $\xi = L$:

$$\tan(m_n L) = -\frac{m_b}{m_n} \quad [\text{D}] \quad (58)$$

6.2 Dimensionless Form

With $x_n = m_n L$ and $b = m_b L$:

$$\tan(x_n) = -\frac{b}{x_n} \quad [\text{D}] \quad (59)$$

6.3 b as Function of EDC Parameters

Substituting $b = \lambda \sigma L^2 / \bar{M}_{\text{Pl}}^2$:

$$\tan(x_n) = -\frac{\lambda \sigma L^2}{\bar{M}_{\text{Pl}}^2 x_n} \quad [\text{Dc}] \quad (60)$$

6.4 Gap Formula

The gap is:

$$m_{\text{gap}} = \frac{x_1(b)}{L} = \frac{x_1(\lambda \sigma L^2 / \bar{M}_{\text{Pl}}^2)}{L} \quad (61)$$

6.5 Limiting Cases

6.5.1 Weak Brane Coupling ($b \rightarrow 0$)

When $\sigma L^2/\bar{M}_{\text{Pl}}^2 \rightarrow 0$ or $\lambda \rightarrow 0$:

$$x_1 \rightarrow \pi, \quad m_{\text{gap}} \rightarrow \frac{\pi}{L} \quad (\text{Neumann limit}) \quad (62)$$

The small- b expansion gives:

$$x_1(b) = \pi - \frac{b}{\pi} + O(b^2) \quad (63)$$

Therefore:

$$m_{\text{gap}} = \frac{\pi}{L} \left(1 - \frac{b}{\pi^2} + O(b^2) \right) \quad (64)$$

6.5.2 Strong Brane Coupling ($b \rightarrow \infty$)

When $\lambda \sigma L^2/\bar{M}_{\text{Pl}}^2 \rightarrow \infty$:

$$x_1 \rightarrow \frac{\pi}{2}, \quad m_{\text{gap}} \rightarrow \frac{\pi}{2L} \quad (\text{Dirichlet limit}) \quad (65)$$

The large- b asymptotic is:

$$x_1(b) = \frac{\pi}{2} + \frac{\pi}{2b} + O(b^{-2}) \quad (66)$$

Therefore:

$$m_{\text{gap}} = \frac{\pi}{2L} \left(1 + \frac{1}{b} + O(b^{-2}) \right) \quad (67)$$

6.5.3 Transition Region

The transition from Neumann to Dirichlet behavior occurs around $b \sim 1$:

$$b = 1 : \quad x_1 \approx 2.80, \quad \frac{x_1}{\pi} \approx 0.89 \quad (68)$$

6.6 Where Does the Gap Close?

The gap becomes fixed (**[D]** or **[Dc]**) when:

1. λ is determined (e.g., by topological quantization)
2. σ is EDC-derivable (currently **[BL]** or **[Dc]**)
3. L is determined (currently **[I]**+**[BL]**)

Table 4: Gap closure requirements.

If...	Then b is...	Then m_{gap} is...	Status
All three fixed	Fixed	[Dc]	Best achievable
λ free, others fixed	1-param family	[Dc]	Good
L from M_Z	Fixed via identification	[I] + [BL]	Current
Nothing fixed	Free	[OPEN]	Worst

7 Numerical Analysis

7.1 Root-Finding for $x_1(b)$

The first root x_1 of $\tan(x) + b/x = 0$ is computed numerically for $b \in [10^{-3}, 10^3]$.

7.2 Results

Table 5: First root x_1 for various $b = m_b L$.

b	x_1	x_1/π	Limit
0	3.1416	1.000	Neumann
0.01	3.1384	0.999	
0.1	3.1094	0.990	
1	2.7984	0.891	
10	1.7434	0.555	
100	1.5867	0.505	
1000	1.5724	0.500	Dirichlet
∞	1.5708	0.500	

7.3 Discrete λ Demonstration

For $\lambda = \pi n$ with $n = 1, 2, \dots, 10$ and a reference value $\sigma L^2/\bar{M}_{\text{Pl}}^2 = 1$:

Table 6: Gap for discrete $\lambda = \pi n$ (demonstration only).

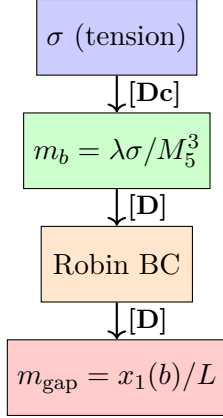
n	$\lambda = \pi n$	$b = \lambda \cdot 1$	$x_1(b)$
1	3.14	3.14	2.46
2	6.28	6.28	1.90
3	9.42	9.42	1.76
4	12.57	12.57	1.69
5	15.71	15.71	1.65
6	18.85	18.85	1.62
7	21.99	21.99	1.61
8	25.13	25.13	1.60
9	28.27	28.27	1.59
10	31.42	31.42	1.58

Note: This table demonstrates parameter dependence, not physical prediction. The actual value of $\sigma L^2/\bar{M}_{\text{Pl}}^2$ depends on L , which is currently **[I]+[BL]**.

8 Graphical Summary

9 Comparison with Electroweak Scale

Note: This section is *comparison only*. The electroweak identification is **[I]+[BL]**, not a derivation.



(A) $\sigma \rightarrow m_b \rightarrow \text{gap}$

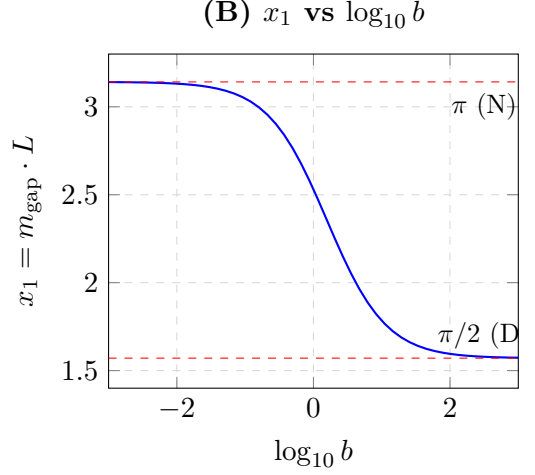


Figure 1: (A) Schematic showing how brane tension σ generates the mass gap through $m_b = \lambda\sigma/M_5^3$ and Robin boundary conditions. Tags indicate derivation status. (B) First eigenvalue x_1 as a function of $\log_{10} b$, transitioning from Neumann limit ($x_1 = \pi$) to Dirichlet limit ($x_1 = \pi/2$).

9.1 Current Identification

In the BLOCK-003 program, the gap is identified with M_Z :

$$m_{\text{gap}} = M_Z = 91.19 \text{ GeV} \quad [\text{I}] + [\text{BL}] \quad (69)$$

This implies:

$$L = \frac{\pi \hbar c}{M_Z} = 6.80 \times 10^{-18} \text{ m} \quad [\text{I}] + [\text{BL}] \quad (70)$$

9.2 Implied b Value

Using this L and the bridge relation:

$$M_5 = \left(\frac{\bar{M}_{\text{Pl}}^2}{L} \right)^{1/3} = 5.56 \times 10^{12} \text{ GeV} \quad (71)$$

The brane mass is:

$$m_b = \frac{x_1}{L} - \frac{\pi}{L} \cdot (\text{correction from Robin}) \quad (72)$$

If the gap equals exactly π/L (Neumann), then $m_b = 0$. If there is a Robin correction:

$$m_{\text{gap}} = \frac{x_1(b)}{L} < \frac{\pi}{L} \quad (73)$$

9.3 Matching Condition

For the gap to equal M_Z with a Robin BC:

$$\frac{x_1(b)}{L} = M_Z \quad (74)$$

This requires:

$$x_1(b) = M_Z \cdot L = M_Z \cdot \frac{\pi \hbar c}{M'_Z} = \pi \cdot \frac{M_Z}{M'_Z} \quad (75)$$

If $M'_Z = M_Z$ (self-consistent identification), then $x_1 = \pi$ (Neumann).

For Robin with $b > 0$, we need $x_1 < \pi$, implying $m_{\text{gap}} < \pi/L$, which would give a gap smaller than the Neumann value.

Conclusion: The pure Neumann case ($m_b = 0$) is compatible with $m_{\text{gap}} = M_Z$ if $L = \pi\hbar c/M_Z$. Robin corrections would modify this.

10 Epistemic Ledger

10.1 What Is Fixed

Table 7: Fixed quantities in the m_b - σ relation.

Quantity	Expression	Status	Tag
Dimensional scaling	$m_b \propto \sigma/M_5^3$	Derived	[D]
Bridge relation	$\bar{M}_{\text{Pl}}^2 = M_5^3 L$	Derived	[D]
Spectral equation	$\tan(x_n) = -b/x_n$	Derived	[D]
Gap bounds	$\pi/2L < m_{\text{gap}} < \pi/L$	Derived	[D]
$b = m_b L$	$b = \lambda \sigma L^2 / \bar{M}_{\text{Pl}}^2$	Derived with λ	[Dc]

10.2 What Is Open

Table 8: Open quantities requiring further input.

Quantity	Current Status	Required Input	Tag
λ	Undetermined	Topological constraint?	[OPEN]
σ	External or EDC-derivable	EDC dynamics?	[BL] or [Dc]
L	Identified with $\pi\hbar c/M_Z$	Internal derivation?	[I]+[BL]
m_{gap}	Identified with M_Z	Above closures needed	[I]+[BL]

10.3 Path to GDC-2 (Calibrated Derivability)

To achieve [Dc] for the gap:

1. Fix λ (e.g., $\lambda = \pi$ from topological argument) [P]→[D]
2. Express σ in terms of $\bar{M}_{\text{Pl}}$ (if possible) [BL]→[Dc]
3. Express L in terms of $\bar{M}_{\text{Pl}}$ (currently NO-GO from v16)

10.4 Path to GDC-1 (Full Derivability)

To achieve [D] for the gap:

1. All of GDC-2 requirements
2. No external mass scale (even $\bar{M}_{\text{Pl}}$) in the final expression

This is extremely challenging and likely impossible within current EDC.

11 Conclusions

11.1 Summary of Results

1. **Dimensional relation established:**

$$m_b = \lambda \frac{\sigma}{M_5^3} \quad [\mathbf{Dc}] \quad (76)$$

2. **Dimensionless control parameter:**

$$b = m_b L = \lambda \frac{\sigma L^2}{\bar{M}_{\text{Pl}}^2} \quad [\mathbf{Dc}] \quad (77)$$

3. **Topological pinning candidate:** $\lambda = \pi n$ or $2\pi n$ **[P]**
4. **Connection to v26 spectrum:** $\tan(x_n) = -b/x_n$ with $b = \lambda \sigma L^2 / \bar{M}_{\text{Pl}}^2$
5. **Gap status:** Remains **[I]**+**[BL]** until λ and L are fixed

11.2 What This Note Achieves

- Maps how m_b arises from σ (dimensional + action-level)
- Identifies λ as the key undetermined coefficient
- Proposes topological quantization as a candidate mechanism
- Shows where the gap “closes” in parameter space

11.3 What This Note Does NOT Achieve

- Does not determine λ
- Does not derive L from EDC-internal principles
- Does not upgrade gap to **[D]** or **[Dc]**

11.4 Next Steps (for P36+)

1. Investigate specific topological mechanisms that could fix λ
2. Explore whether σ can be expressed in terms of α or other EDC-derivables
3. Consider warped geometry where L and k might be related

A Detailed Dimensional Analysis

A.1 All Possible Combinations

With available quantities σ , M_5 , L , $\bar{M}_{\text{Pl}}$, we list all combinations giving dimension $[M^1]$:

$$\frac{\sigma}{M_5^3} : \quad M^4/M^3 = M \quad (78)$$

$$\sigma L^3 : \quad M^4 \cdot M^{-3} = M \quad (79)$$

$$\frac{\bar{M}_{\text{Pl}}^2}{L^2 M_5} : \quad M^2 \cdot M^2/M = M \quad (80)$$

$$M_5 : \quad M \quad (81)$$

$$\bar{M}_{\text{Pl}} : \quad M \quad (82)$$

$$\frac{1}{L} : \quad M \quad (83)$$

A.2 Using the Bridge Relation

With $\bar{M}_{\text{Pl}}^2 = M_5^3 L$, we can express M_5 and L in terms of each other:

$$M_5 = \left(\frac{\bar{M}_{\text{Pl}}^2}{L} \right)^{1/3} \quad (84)$$

$$L = \frac{\bar{M}_{\text{Pl}}^2}{M_5^3} \quad (85)$$

Substituting eq. (85) into σ/M_5^3 :

$$m_b = \lambda \frac{\sigma}{M_5^3} = \lambda \frac{\sigma L}{\bar{M}_{\text{Pl}}^2} \quad (86)$$

A.3 Alternative Representations of b

The control parameter $b = m_b L$ can be written:

$$b = \lambda \frac{\sigma L^2}{\bar{M}_{\text{Pl}}^2} = \lambda \frac{\sigma L}{M_5^3} = \lambda \frac{\sigma \bar{M}_{\text{Pl}}^2}{M_5^6} \quad (87)$$

Each form emphasizes different physical content:

$$\frac{\sigma L^2}{\bar{M}_{\text{Pl}}^2} : \quad \text{brane tension relative to 4D gravity at scale } L \quad (88)$$

$$\frac{\sigma L}{M_5^3} : \quad \text{brane-induced coupling in 5D gravity} \quad (89)$$

$$\frac{\sigma \bar{M}_{\text{Pl}}^2}{M_5^6} : \quad \text{hierarchy between 4D and 5D gravity} \quad (90)$$

The combination σ/M_5^3 is preferred because:

- It involves only brane (σ) and bulk (M_5) scales
- It does not require knowing L a priori
- It arises naturally from the gravitational coupling

B Action Derivation Details

B.1 Brane-Bulk Coupling

The coupling between bulk field Φ and brane tension can be modeled as:

$$S_{\text{int}} = - \int d^4x \sqrt{-\gamma} \sigma(\Phi) \delta(\xi - L) \quad (91)$$

where $\sigma(\Phi) = \sigma_0 + \sigma_1 \Phi + \frac{1}{2} \sigma_2 \Phi^2 + \dots$

B.2 Quadratic Coupling

The quadratic term gives:

$$S_{\text{quad}} = -\frac{1}{2} \int d^4x \sqrt{-\gamma} \sigma_2 \Phi^2 \big|_{\xi=L} \quad (92)$$

Identifying $m_b = \sigma_2$, we need $[\sigma_2] = M$.

If $\sigma_2 \propto \sigma/M_5^3$, we recover eq. (9).

B.3 Junction Condition Derivation

From the Israel junction conditions with Φ -dependent tension:

$$[K] = -\kappa_5^2 \sigma(\Phi) \quad (93)$$

Expanding:

$$[K] = -\kappa_5^2 \sigma_0 - \kappa_5^2 \frac{d\sigma}{d\Phi} \Phi - \frac{1}{2} \kappa_5^2 \frac{d^2\sigma}{d\Phi^2} \Phi^2 + \dots \quad (94)$$

The Φ^2 term contributes to the effective brane mass.

C Numerical Implementation

C.1 Root-Finding Algorithm

The transcendental equation $f(x) = \tan(x) + b/x = 0$ is solved using:

1. Bracket the first root in $(\pi/2, \pi)$
2. Apply Brent's method for robust convergence
3. Verify residual $|f(x_1)| < 10^{-12}$

The function to solve:

$$f(x) = \tan(x) + \frac{b}{x} = 0 \quad (95)$$

Its derivative for Newton refinement:

$$f'(x) = \sec^2(x) - \frac{b}{x^2} \quad (96)$$

C.2 Asymptotic Approximations

For small b , the perturbation expansion:

$$x_1 = \pi - \frac{b}{\pi} - \frac{b^2}{\pi^3} + O(b^3) \quad (97)$$

For large b , the asymptotic series:

$$x_1 = \frac{\pi}{2} + \frac{\pi}{2b} - \frac{\pi^3}{8b^3} + O(b^{-5}) \quad (98)$$

C.3 Error Bounds

The residual after root-finding:

$$\epsilon = |f(x_1)| = \left| \tan(x_1) + \frac{b}{x_1} \right| < 10^{-12} \quad (99)$$

C.4 Python Code Reference

See `recompute.py` for:

- `find_root(b)`: Returns x_1 for given b
- `scan_b(b_range)`: Scans b and returns (b, x_1) pairs
- `discrete_lambda(n_range)`: Computes for $\lambda = \pi n$